

# Xiaoxiao YU

Agent Engineer | Full-Stack Engineer

[jupitexiaoxiaoyu@gmail.com](mailto:jupitexiaoxiaoyu@gmail.com) | [github.com/JupiterXiaoxiaoYu](https://github.com/JupiterXiaoxiaoYu) | [Portfolio](#) | [LinkedIn](#)

Agent engineer who builds the whole product, from the React frontend to the agent runtime underneath. At Vicking (full-time since July 2026), sole engineer on the agent products that **doubled video-production efficiency**. Full-stack since 2021: Python, TypeScript and Rust backends with React and React Native frontends, in products with thousands of users. Founded a browser-agent startup accepted into Ant Group's MaShang AI Accelerator, and wrote an open-source multi-agent harness with 175 GitHub stars.

## Experience

### Vicking (Beijing Weixi Network Technology)

Jul 2026 – Present

*Agent Engineer, full-time | sole engineer, frontend to agent runtime, on four internal AI production products*

- Versus manual editing, **production efficiency doubled**; the tools serve dozens of internal users across multiple business lines.
- **WeiChuang Drama Studio** (React 19, FastAPI, Temporal, PostgreSQL): a 1-minute episode goes **from script to finished video in 15–30 min**, with episodes in parallel; **~90% go straight to ad placement and delivery**, the rest need only light edits. If a model call fails mid-batch, only the failed shots rerun, so finished shots are never regenerated or paid for twice. Bad generations fall back to another model or get their voice repaired, and every model call is logged so a slow or stuck job can be traced to the exact call. ~5,000 tests.
- **Weixi Studio** (TypeScript, React, Remotion, PostgreSQL, MCP): turned OpenChatCut, a local-first single-user agentic video editor, into a **multi-tenant SaaS** running several production lines (drama remix, social re-cuts, packaging, digital-human narration, AI pre-roll and recreation) that share one asset library, template-driven batch production, semantic search and cost reporting. Agent tasks resume after a worker crash; paid or irreversible tool calls are never retried blindly, and unclear outcomes go to a person. Over MCP, outside agents and editors work on one timeline.
- **Screenplay Agent Studio** (FastAPI, Pi agent SDK, OpenTelemetry): built an eval comparing one lead agent with a lead plus specialists; the specialists cost **3.7× the tokens for no quality gain**, so each of the 6 writing stages runs a single lead. Drafts live in files rather than chat history, so long sessions can be compressed without losing work; writers approve every edit; alerts fire when agent runs start failing or a decision waits too long.
- **Creative Engine** (TanStack Start, pgvector): a writing agent that drafts from analyzed reference videos and keeps what it observed apart from what it inferred; ran 14 controlled experiments with an LLM judge and removed every prompt technique that did not measurably help.
- **Clip pipeline (team project)**: ported Weixi Studio's AI pre-roll, scene recreation, watermark removal and stitching into the team's v2 pipeline, which turns dramas into ad clips at scale, and built the unattended uploader that ships them to the ad platform, work previously done by hand, with no duplicate submissions. Separately built a GPU redaction pipeline (fine-tuned RF-DETR) that masks sensitive objects at **2× real time**, ~80% accurate.

### Delphinus Lab

Nov 2024 – Jul 2026

*Developer Relations Engineer → Ecosystem Director (Jun 2025) | Remote, concurrent with MPhil*

- Rebuilt the zkWasm browser SDK from scratch so apps use one wallet connection across both chain layers; wrote the CLI that scaffolds new apps; maintained the TypeScript services that settle deposits and withdrawals.
- Solo-built staking, launchpad, prediction-market and social-binder apps end to end (Rust application logic, TypeScript services, React frontends), plus the zkWasm Hub site and developer docs; ran DevOps for 8+ ecosystem apps, which reached **thousands of users** and **multi-million-USD TVL**.
- Led the team that built a hosted Kubernetes platform where developers deploy straight from GitHub and watch build status and logs live; it hosts **10+ projects**.

### Earlier roles (2021–2024, alongside undergraduate studies)

- **Rosen** — Founding Full-Stack Engineer (2022–23): React / React Native apps for iOS, Android and Web; backed by PayPal and Circle. **2PM.Network** — Co-Founder & CEO (2024). **HashPunk** — Product Manager (2021–22), raised US\$1M. **1CIRCLE** — Full-Stack Engineer (2021–22). **Conflux** — Intern (2021).

## Selected Projects

### Resona — The Launch Layer for AI-Built Products

May – Jun 2026

*Founder & CEO, team of three; built the entire product myself | Ant Group MaShang AI Accelerator; trial customers*

- For people who build products with AI: paste a URL and one sentence about what to show, and **about 30 minutes** later get a launch kit (video, social cuts, screenshots, GIFs, copy) made from the real running product, not a mock-up.
- **GUI-agent exploration**: agents operate the live product in parallel cloud sandboxes (Stagehand, Playwright), get past login walls and unpredictable UI states, and judge which path is worth showing; risky actions are gated.
- **Compositional rendering**: screen recordings, AI-generated code animation (GSAP, Remotion) and AI voiceover are cut into one narrative and exported in several formats; ~500 tests.

- Designed 17 role agents and 25 skills that take a novel through ingest, analysis, episode planning, writing, review and storyboard / video prompts. A review director gates every episode, and failures loop back for rewrite and re-review.
- Progress is saved to files, so a run can stop and resume at any episode without the context window filling up; hybrid ChromaDB + TF-IDF search pulls similar reference scripts to ground the writing.

ShotLoom — AI Film Shot-Breakdown Workspace

Open source, sole author | FastAPI, Redis/RQ, React, TypeScript, Qwen vision + ASR

- Built a shot-by-shot film breakdown: scene detection and clip cutting, then Qwen vision analysis of shot size, camera movement, lighting and narrative function, aligned with ASR transcripts and a whole-film continuity report.
- Long films are split into overlapping chunks so no scene is lost at a boundary; progress streams live and survives a page refresh; Excel/PDF export; 107 end-to-end tests.

RingChat — Agent-Native Messaging

Jan – Mar 2026

Team lead, client engineer | Kotlin / Jetpack Compose, Web, iOS | OpenClaw (self-hosted agent gateway)

- Led a **6-person team** building a cross-platform IM in which every user gets a personal AI agent, and worked mainly on the clients (now continued by another team as Gold House). On Android, built the chat UI, optimistic media upload, contact and group search, unread badges and multi-device sync.

Skills

|                   |                                                                                                                                                                  |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Agents            | Context engineering, tool and MCP design, multi-agent orchestration, browser agents (Stagehand, Playwright, AgentBay), human-in-the-loop, evals and LLM-as-judge |
| Backend           | Python (FastAPI, Celery), TypeScript / Node.js, Rust, C# / .NET; Temporal workflows, PostgreSQL / pgvector, Redis, BullMQ, SSE, WebSocket                        |
| Frontend & mobile | TypeScript, React 19, TanStack, Next.js, Remotion; React Native, Kotlin (Compose)                                                                                |
| ML & media        | PyTorch, RF-DETR fine-tuning, TensorRT, FFmpeg; OpenAI, Anthropic, Gemini, Qwen, Seedance                                                                        |
| Infra & other     | Docker, Kubernetes / Helm, CI/CD, OpenTelemetry; Solidity / Foundry, C++ (ESP32, LVGL)                                                                           |

Education

The Hong Kong University of Science and Technology (Guangzhou)

Sep 2024 – Oct 2026

Red Bird MPhil in Data Science and Analytics | graduated Oct 2026

The University of Edinburgh

Sep 2020 – Jun 2024

MA in Cognitive Science, First Class Honours | Exchange: Fudan University, Computer Science (2022–23)

Publication

Xiaoxiao YU, L. Du, C. Zeng, L. Deng, H. Luo, X. Gao, J. Tang. *SocialAttributionQA: A Benchmark for Attribution-Evidence-Aware Social Web Question Answering*. **WISE 2026** (CCF B), accepted; Springer LNCS. First author: built a 1,200-question benchmark over 725k Mastodon posts and the ATLAS graph-plus-retrieval solver (**95.3%** accuracy), with ablations and error analysis.

Selected Hackathons & Honors

33 hackathon and competition awards (12 first place or champion); AI and full-stack highlights:

Ava Box — Voice-First Trading Agent

AVE CLAW Hackathon 2026 — Champion

Sole developer | ESP32-S3, C++, Python, LVGL

- Built a handheld voice agent that routes speech to trading tools based on the current screen; every order needs a button confirmation, and the LVGL UI is shared between firmware and a desktop simulator. Extracted into DeviceKit, an open framework where the chain, speech and language models can be swapped, with over-the-air updates.

OutcomeX — AI Delivery & Settlement

HashKey Chain Horizon 2026 — 2nd, DeFi Track

Sole developer | FastAPI, AgentSkillOS, Solidity

- Built a marketplace where buyers pay for an AI deliverable: an agent runtime (AgentSkillOS) does the work with live progress, and payment stays in escrow until the buyer accepts, with refunds and revenue sharing on-chain.

|      |                                                       |      |                                                        |
|------|-------------------------------------------------------|------|--------------------------------------------------------|
| 2025 | Zama Bounty, Universal FHEVM SDK — <b>Winner</b>      | 2024 | BeWater AI Crypto — <b>1st, Security &amp; Privacy</b> |
| 2025 | ICBC Cup FinTech, Guangdong — <b>First Prize</b>      | 2024 | ETH Beijing — <b>1st, Scroll Track</b>                 |
| 2024 | ETH Shanghai — <b>Overall Champion</b>                | 2024 | NEAR Hackathon — <b>ZettaBlock AI Bounty (solo)</b>    |
| 2024 | Shenzhen Intl. FinTechathon — <b>2nd, Blockchain</b>  | 2022 | Algorand Green House — <b>3× Champion (solo)</b>       |
| 2024 | MEME.ZKWASM() Hackathon — <b>2nd, Game (solo)</b>     | 2022 | YRDA Data Open Innovation — <b>2nd Prize</b>           |
| 2023 | Cosmos Developer Hackathon — <b>1st Place</b>         | 2022 | OpenMPC Privacy Computing — <b>Winning Prize</b>       |
| 2023 | DAO-Fi Fusion — <b>2nd, Best Technical (solo)</b>     | 2021 | PaddlePaddle Anti-Fraud ML — <b>1st (solo)</b>         |
| 2021 | Reach on Conflux Bounty Hack — <b>Champion (solo)</b> | 2021 | Huawei Cloud Cup AI Contest — <b>Winning Prize</b>     |

Founded LambdaSoc (Edinburgh) and Lambda Web3 (**1,000+ members**); **7,000+** Xiaohongshu followers for engineering content. Languages: Chinese (native), English (professional).